ME2201: Kinematics and Dynamics of Machinery

Tutorial 1

Submission Date: 05-Aug-2026, in class

1. Draw the kinematic diagram of the crusher shown in Figure 1. Identify the types of kinematic pairs. How many degrees of freedom does it have? Is it a mechanism?

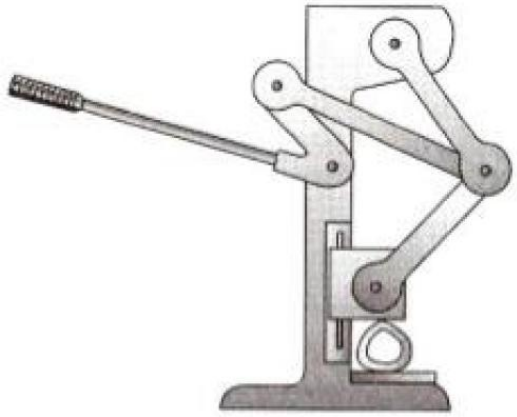


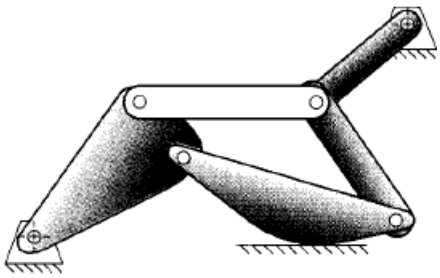
Figure 1: Crusher

2. Draw the kinematic diagram of the front loader attachment of the backhoe loader, JCB 3DX variant (include the piston-cylinder assemblies as a part of the mechanism). Identify the types of joints. Calculate the degrees of freedom for this mechanism. You may watch videos on the internet to understand the operation of the loader.

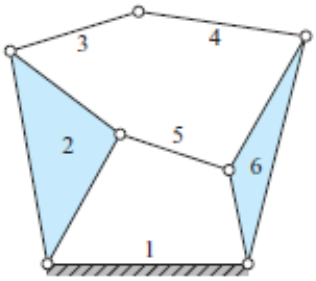


Figure 2: Loader attachment (https://www.jcb.com/en-IN/)

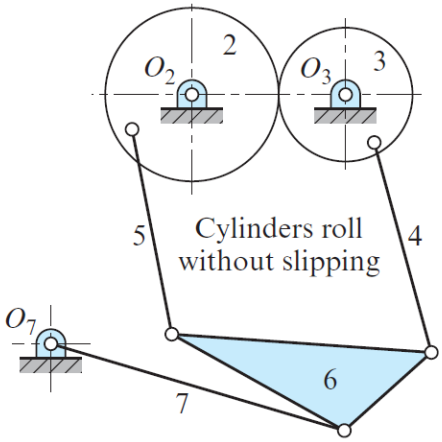
3. (Optional) Calculate the mobility for the mechanisms shown in Figure 3. Are there any redundant degrees of freedom in the mechanisms shown? Are there any structures in the figures shown?



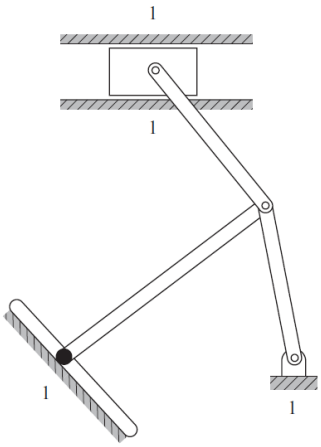
3. 1



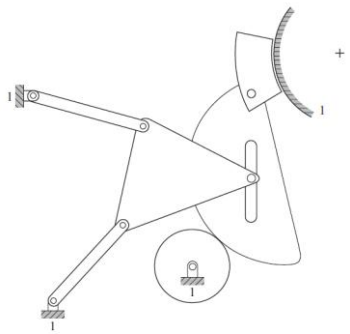
3. 2



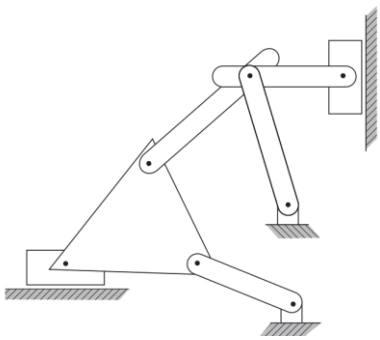
3. 3



3. 4



3. 5



3. 6

Figure 3